Chest X-ray Thoracic Pathology Classification

Technical Report

Contents

1 Introduction

Problem: Automated classification of chest X-ray images into 20 thoracic pathology categories (e.g., Atelectasis, Cardiomegaly, Effusion, Pneumothorax, No Finding, etc.). Misdiagnosis of pathologies carries significant clinical cost — a missed finding (false negative) is far more dangerous than a false alarm (false positive).

Why it matters: Radiologists handle thousands of X-rays daily, and automated screening can prioritise urgent cases. The competition's asymmetric scoring metric (FN penalty = 5× FP penalty) directly reflects this clinical reality: missing a disease is penalised much more heavily than flagging a healthy patient.

Dataset: The competition provides labelled chest X-ray images with one-hot encoded multi-class labels across 20 categories. Each image belongs to exactly one class (single-label classification). The dataset exhibits severe class imbalance — "No Finding" dominates while rare pathologies like Hernia and Pneumothorax have very few samples.

Objective: Build a classifier that maximises the macro-averaged asymmetric cost score on the held-out test set, where the score per class is defined as:

$$S_c = \frac{TP_c - FP_c - 5 \cdot FN_c}{N_c} \tag{1}$$

2 Methodology

2.1 Model Architecture

DenseNet-121 was chosen as the backbone for the following reasons:

- **Feature reuse:** Dense connections allow each layer to access feature maps from all preceding layers, which is particularly effective for medical imaging where subtle features matter.
- **Parameter efficiency:** Compared to ResNet, DenseNet achieves comparable accuracy with fewer parameters, which helps prevent overfitting on a moderately-sized medical dataset.
- **Proven medical imaging backbone:** DenseNet-121 has been widely used in CheXpert, NIH ChestX-ray14, and similar benchmarks.

A custom classifier head replaces the default head:

```
BatchNorm1d -> Dropout(0.4) -> Linear(1024 -> 512) -> ReLU -> BatchNorm1d -> Dropout
    (0.2) -> Linear(512 -> 20)
```

The two-layer head with batch normalization and graduated dropout ($0.4 \to 0.2$) provides regularisation while allowing sufficient capacity for 20-class discrimination.

2.2 Preprocessing

Step	Detail
Resize	All images resized to 256×256 pixels.
Normalisation	ImageNet mean/std (`[0.485, 0.456, 0.406]` / `[0.229, 0.224, 0.225]`) to match pretrained backbone.
Colour mode	Converted to RGB (3-channel).

Table 1: Image Preprocessing Steps

2.3 Data Augmentation

Training-time augmentations simulate realistic clinical variations:

Augmentation	**Parameters**	**Rationale**
Random Horizontal Flip	$p = 0.5$	X-rays can have laterality variations.
Random Rotation	$\pm 10^\circ$	Slight patient positioning differences.
Colour Jitter	brightness=0.2, contrast=0.2	Different X-ray machine settings.
Random Affine	translate=5%, scale=95–105%	Minor alignment/zoom differences.

Table 2: Training-Time Data Augmentations

2.4 Handling Class Imbalance

A three-pronged strategy was used:

1. **Sqrt-inverse-frequency weighted sampling:** `WeightedRandomSampler` with weights = $1/\sqrt{\text{class\_count}}$. The square root provides a gentler balancing than full inverse-frequency — it boosts rare classes without completely suppressing the majority class ("No Finding").

2. **Balanced Cross-Entropy loss with label smoothing:** Class weights computed as $\sqrt{\text{total}/(\text{num\_classes} \times \text{freq})}$, normalised so weights sum to num_classes. Label smoothing ($\epsilon = 0.05$) prevents overconfidence and improves calibration.

3. **Cost-sensitive prediction at inference (not training):** Rather than baking the asymmetric FN penalty into training (which would distort learned representations), the model is trained for balanced accuracy and the competition's cost structure is applied only at prediction time via a tunable α parameter.

2.5 Training Strategy

- **Transfer learning:** Initialised from ImageNet-pretrained DenseNet-121 weights.
- **Optimiser:** AdamW (lr = 3e-4, weight_decay = 1e-4).
- **Scheduler:** Cosine annealing (T_{max} = 12 epochs, η_{min} = 1e-6) for smooth learning rate decay.
- **Mixed precision:** FP16 via `torch.cuda.amp` for faster training and reduced memory.
- **Early stopping:** Patience of 6 epochs on validation competition score.

2.6 Stratified K-Fold Cross-Validation

5-fold stratified split ensures each fold preserves class distribution. 3 out of 5 folds are trained (folds 0, 1, 2) to build a diverse ensemble while keeping compute reasonable.

2.7 Inference Pipeline

- **Multi-model ensemble:** Predictions averaged across 3 fold models.
- **Test-Time Augmentation (TTA):** 5 rounds of random horizontal flip + slight rotation per model. Each test sample gets $3 \times (1 + 5) = 18$ forward passes, and probabilities are averaged.

- **Cost-sensitive threshold tuning:** An α parameter (searched from 0.0 to 1.0 in steps of 0.05) controls how strongly to adjust predictions by inverse class frequency. Alpha is tuned on the last fold's validation set to maximise the competition score before being applied to test predictions.

3 Experiments and Results

3.1 Experiments Performed

Experiment	Description
Baseline training	DenseNet-121 with standard CE loss, no augmentation.
Class balancing	Added sqrt-inverse-frequency weighted sampling + balanced CE loss.
Data augmentation	Added flip, rotation, jitter, affine transforms.
Label smoothing	$\epsilon = 0.05$ to improve probability calibration.
Fold ensemble	3-fold ensemble vs. single model.
TTA	5-round TTA to reduce prediction variance.
Cost-sensitive alpha tuning	Searched for optimal α on validation set.

Table 3: Summary of Experiments Performed

3.2 Hyperparameter Variations

Parameter	Values Tried	Final Value
Image size	224, 256, 384	**256** (best speed–accuracy trade-off)
Batch size	32, 64, 128	**64**
Learning rate	1e-3, 3e-4, 1e-4	**3e-4**
Dropout	0.3, 0.4, 0.5	**0.4**
Num epochs	10, 12, 15	**12** (early stopping often triggers at $\sim$10)
FN weight	3.0, 5.0, 7.0	**5.0** (matches competition)
Alpha	0.0 – 1.0 (step 0.05)	Tuned per run

Table 4: Hyperparameter Search Space and Final Selections

3.3 Results

Model / Configuration	Weighted F1 Score
DenseNet-121 (single fold, no TTA)	~ -5.25
DenseNet-121 (3-fold ensemble + TTA)	~ -4.82
+ Cost-sensitive alpha tuning	**Best validation score**

Table 5: Model Performance Results

3.4 Validation Strategy

- **Stratified K-Fold (5 splits):** Ensured each fold maintained the same class distribution as the full training set.

- **Competition metric tracking:** Monitored the asymmetric cost score (not just accuracy or F1) throughout training.
- **Confusion matrix analysis:** Identified which pathologies were most commonly confused (e.g., Atelectasis ↔ Infiltration).
- **Classification report:** Per-class precision, recall, and F1 for all 20 classes.

4 Conclusion

Summary of Approach

A DenseNet-121 model fine-tuned with transfer learning from ImageNet, trained with balanced CE loss and sqrt-inverse-frequency sampling, and enhanced at inference time with 3-fold ensembling, 5-round TTA, and cost-sensitive α-tuned prediction.

Key Learnings

1. **Decouple training and scoring objectives:** Training with symmetric balanced loss and applying asymmetric costs only at inference gave better results than baking FN penalties into training, which caused the model to ignore majority classes.
2. **Sqrt-balancing > full inverse-frequency:** Full inverse-frequency sampling/weighting was too aggressive — it suppressed "No Finding" to near-zero predictions. The square root variant provides a much gentler and more effective balance.
3. **Ensemble + TTA is highly effective:** Averaging predictions across 3 folds and 5 TTA rounds significantly reduced prediction noise and improved the competition score.
4. **Alpha tuning matters:** The optimal α varies between runs, confirming that a one-size-fits-all cost adjustment is suboptimal. Grid search on validation data is essential.
5. **Medical imaging benefits from conservative augmentation:** Unlike natural image tasks, chest X-rays have limited variation, so aggressive augmentations (large rotations, heavy colour shifts) hurt performance.

5 References

References

[1] Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). *Densely Connected Convolutional Networks.* CVPR.

[2] Rajpurkar, P., et al. (2017). *CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning.* arXiv:1711.05225.

[3] Irvin, J., et al. (2019). *CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison.* AAAI.

[4] PyTorch documentation — `torchvision.models`, `torch.cuda.amp`.

[5] He, K., et al. (2016). *Deep Residual Learning for Image Recognition.* CVPR.

[6] Loshchilov, I., & Hutter, F. (2019). *Decoupled Weight Decay Regularization.* ICLR.